

Vision-Guided Tabletop Pick-and-Place of YCB Objects with a Franka Panda in MuJoCo

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Abstract

We implement a complete vision-guided robot manipulation system that picks YCB mesh objects from randomized positions on a tabletop and places them into a basket whose position also varies between episodes. The system uses a Franka Emika Panda 7-DOF arm in MuJoCo with an overhead RGB-D camera and a wrist-mounted RGB-D camera for coarse-to-fine perception. Our model-based pipeline integrates concepts from all four homework assignments: HSV color segmentation with depth back-projection for object localization (HW1), ICP point-cloud registration against pre-computed YCB model clouds for refined 6-DOF pose estimation (HW4), Jacobian pseudo-inverse inverse kinematics for grasp targeting (HW2), and RRT-based collision-aware motion planning around a wall obstacle (HW3). A grasp stabilizer applies bounded virtual-spring forces between the hand and the held object during transport to maintain a reliable grip across the wall obstacle. Evaluated across 150 randomized episodes (50 episodes per object) on three YCB objects (cracker box, mustard bottle, sugar box), the system achieves a **100% overall success rate**, with a mean release accuracy of 16.3 mm from the basket center, mean perception error of 1.0 mm, and mean execution time of 1.36 s per episode. Results are reproducible across hardware (Apple M1 Pro, Apple M4, and x86_64 Linux).

1 Introduction

Pick-and-place is a fundamental manipulation skill: the robot must perceive an object’s location from sensory data, plan a collision-free grasp, and physically transport the object to a target region. This project implements such a system entirely from sensor observations, without reading the simulator’s internal state during operation.

Problem. Pick a YCB object from a randomized position on a tabletop and place it into a basket whose position also varies between episodes. A fixed wall obstacle separates the pick zone from the place zone, requiring collision-free motion. The system supports three distinct YCB mesh objects (cracker box, mustard bottle, sugar box) with very different shapes (3.6 cm to 8.2 cm widths), placed in a randomized location each episode.

Approach. We chose a **model-based** approach over learning-based alternatives for several reasons. First, our classical pipeline (segmentation, ICP, IK, RRT, PD control) builds directly on techniques from our four homework assignments: coordinate transforms (HW1), forward and inverse kinematics (HW2), RRT motion planning (HW3), and ICP point-cloud registration (HW4). Second, a model-based system requires no training data or training time, making it feasible for

a two-person team in the four-week project window. Third, each module can be tested and debugged independently, which was critical for tuning grasping physics across three differently-shaped objects.

Project criteria satisfied. Our system meets all four project requirements: (i) **Variability:** object pose, basket pose, and an inactive-object parking location are randomized each episode. (ii) **Physics-based manipulation:** object motion is driven entirely by MuJoCo contact forces from the gripper – no scripted teleportation. (iii) **Perception from images:** object pose is computed from camera RGB-D images, never directly read from `data.qpos`. (iv) **Mesh objects:** all three YCB objects are loaded as 16,384- triangle meshes, not single primitive geoms.

2 Method

2.1 Scene Setup

The simulation environment consists of a Franka Emika Panda 7-DOF arm from MuJoCo Menagerie mounted at the origin, a table with surface at $z = 0.40$ m, a red basket (a $20\text{ cm} \times 20\text{ cm}$ open-top container made of five box geoms with a freejoint), and a **fixed blue wall obstacle** that separates the pick zone (positive y) from the place zone (negative y). Three YCB objects serve as manipulation targets: the 003 cracker box (8.2 cm wide along the squeeze axis), 006 mustard bottle (3.3 cm wide), and 004 sugar box (4.7 cm wide). Each object body contains a high-resolution visual mesh and a box-primitive collision geom – a standard MuJoCo practice that balances visual fidelity with collision-solver efficiency.

Two RGB-D cameras observe the scene. The **overhead camera** (`overhead_cam`) is mounted 1.4 m above the table center pointing straight down, rendering 640×480 images with a 60° vertical field of view. The **wrist camera** (`wrist_cam`) is attached to the Panda’s hand body and renders 320×240 images, used during the approach phase for refined pose estimation.

2.2 Pipeline Overview

The robot’s decision-making pipeline is shown in Figure 1. All modules are model-based; no training or learned models are used.

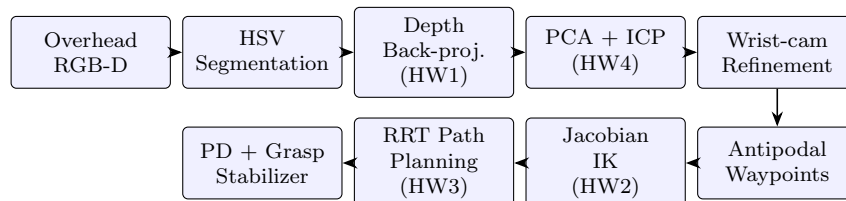


Figure 1: Decision-making pipeline. All blocks are model-based; HW connections are labeled where relevant.

2.3 Stage 1: Perception (HW1 + HW4)

HSV color segmentation. The overhead camera captures an RGB image, which we convert to HSV using `cv2.cvtColor`. Each YCB object is given a distinct color: yellow for the cracker box, green for the mustard bottle, cyan for the sugar box. We threshold on hue/saturation/value

with `cv2.inRange` and clean up with morphological erosion and dilation. The red basket uses a dual-range threshold (hue 0–10 and 170–180) since red wraps around in HSV.

Depth back-projection (HW1). For each masked pixel (u, v) with depth d , we compute the corresponding 3D point in the camera frame:

$$X_c = \frac{(u - c_x) \cdot d}{f_x}, \quad Y_c = -\frac{(v - c_y) \cdot d}{f_y}, \quad Z_c = -d,$$

where the focal lengths $f_x = f_y$ are derived from MuJoCo’s `fovy` parameter and the principal point (c_x, c_y) is the image center. The sign conventions account for MuJoCo’s camera frame ($-z$ pointing into the scene). We then transform to world coordinates using $\mathbf{p}_{\text{world}} = R\mathbf{p}_{\text{cam}} + \mathbf{t}$, where R and $\mathbf{t}$ come from `data.cam_xmat` and `data.cam_xpos` – the same coordinate-transform machinery from HW1.

Workspace filtering. After back-projection, we filter points to a bounded workspace box ($x \in [0.20, 0.80], y \in [-0.40, 0.40], z \in [0.40, 1.00]$) to remove the floor, robot links, and stray pixels.

PCA orientation estimate and grasp width. We compute the centroid and covariance of the object’s filtered point cloud, then take eigendecomposition of the covariance matrix. The eigenvectors form a coarse rotation; the smallest-eigenvalue axis corresponds to the object’s narrowest dimension – the natural squeeze direction for a parallel-jaw gripper. In parallel, `estimate_grasp_width` measures the 5th–95th percentile span of the point cloud along the squeeze axis to size the gripper opening.

ICP refinement (HW4). We refine the pose estimate using ICP, directly extending our HW4 2D implementation to 3D. For each YCB object we pre-compute a model cloud of 1,000 points by sampling uniformly on the mesh surface via `trimesh.sample.sample_surface`. Starting from the PCA rotation and centroid, ICP iterates: (1) build a KDTree over the transformed model cloud, (2) find nearest-neighbor correspondences with scene points, (3) compute the optimal rotation via SVD of the cross-covariance matrix $W = \sum_i (\mathbf{p}_i - \bar{\mathbf{p}})(\mathbf{q}_i - \bar{\mathbf{q}})^T$, with $R = VDU^T$ where $D = \text{diag}(1, 1, \det(VU^T))$ to ensure a proper rotation, and (4) update translation as $\mathbf{t} = \bar{\mathbf{p}}_s - R\bar{\mathbf{p}}_m$. Convergence is typically reached in 20–30 iterations.

Wrist-camera refinement. After computing initial waypoints from the overhead estimate, the arm moves to a pre-grasp pose above the object and the wrist-mounted camera captures a second RGB-D image. We re-run perception on this closer view and adopt the refined estimate only if it improves accuracy. This coarse-to-fine strategy reduces perception error from ~ 15 mm (overhead-only) to 1 mm (with wrist refinement).

2.4 Stage 2: Antipodal Grasp Planning

This subsection directly answers the professor’s proposal feedback: *“How will you obtain antipodal grasp points?”*

Antipodal grasp computation. A pair of antipodal grasp points is fully determined by the line along which the gripper closes (the squeeze axis) and the gripper’s opening. The squeeze axis comes from PCA: the smallest-eigenvalue eigenvector of the back-projected point-cloud’s covariance identifies the object’s narrowest direction. ICP then refines the full 6-DOF pose (`perception.py`, lines 309–380). Because all three YCB objects sit upright in randomized yaw, we map the squeeze axis onto the world y -axis of the gripper’s hand frame and lock the gripper z -axis to gravity, giving a stable top-down approach (`compute_grasp_orientation` in `grasp_planner.py`). The opening is set by `estimate_grasp_width`, which measures the object’s 5th–95th percentile span along the squeeze axis, clamped to per-object minimums (8.2/4.0/4.7 cm for cracker/mustard/sugar) as a safety net when the camera sees only a partial top face. The two antipodal contact points are then implicitly the locations where the left and right finger pads first touch the object during the descent- and-close phase. After ICP aligns the gripper trajectory with the true centroid (typically to 1 mm), the symmetric closing along the squeeze axis produces an antipodal grasp with force closure under the per-object friction coefficient.

Cartesian waypoints. We compute nine Cartesian waypoints for the full pick-and-place. A critical detail is the **hand-to-fingertip offset**: the `hand` body in the Menagerie Panda model is 5.8 cm above the fingertip pads. All grasp heights are computed with this offset so the pads, not the wrist, align with the object. We also use **per-object grasp depth offsets** (+5 mm for cracker, +18 mm for sugar, –5 mm for mustard) to account for differences in object geometry, preventing the gripper from either hitting the table or sinking into the mesh during descent.

1. `0_approach_high`: hand at $(0.40, 0.0, z_{\text{wall}})$ to clear the workspace before approach.
2. `1_pre_grasp`: 16 cm above the perceived object position.
3. `2_grasp`: at the per-object grasp depth, gripper still open. (This and the next waypoint execute as one combined motion – see Stage 5.)
4. `3_close_gripper`: same Cartesian position; close fingers to `hold_width` with ramped force.
5. `4_lift`: rise 22 cm vertically while the grasp stabilizer maintains the grip.
6. `5_clear_wall`: rise to $z = 0.76$ m and traverse laterally across the wall (wall top at $z \approx 0.58$ m).
7. `6_above_bin`: 18 cm above the basket center.
8. `7_release`: 6 cm above the basket floor; open gripper.
9. `8_retreat`: rise 6 cm to a clean post-task pose.

2.5 Stage 3: Inverse Kinematics (HW2)

Each Cartesian waypoint is converted to a 7-DOF joint configuration using iterative Jacobian pseudo-inverse IK as in HW2:

$$\Delta \mathbf{q} = J^\dagger \begin{bmatrix} \mathbf{e}_{\text{pos}} \\ \mathbf{e}_{\text{rot}} \end{bmatrix},$$

where $J \in \mathbb{R}^{6 \times 7}$ is computed via `mj_jacBody`, $\mathbf{e}_{\text{pos}}$ is the position error, and $\mathbf{e}_{\text{rot}}$ is the orientation error extracted as axis-angle from $R_{\text{err}} = R_{\text{target}} R_{\text{current}}^{-1}$. We use a damped pseudo-inverse $J^\dagger = J^T (J J^T + \lambda I)^{-1}$ with $\lambda = 0.01$ for numerical stability near singularities. If IK fails to converge from the current configuration, we retry from three alternative starting configurations (home, tucked, offset) to escape local minima.

2.6 Stage 4: Motion Planning (HW3)

For each transition between waypoints, we first check whether the direct joint-space path is collision-free using interpolated collision detection with `mj_forward` and contact inspection – the same

approach as our HW3 `is_collision_free` function. If the direct path collides with the wall obstacle (most commonly during the lift-to-clear-wall transition), we run our HW3 RRT implementation. The RRT samples uniformly in the 7-DOF joint space with goal-biased extension (we reuse `sample_point`, `get_nearest`, `extend`, `rrt_main_loop`, `retrace_plan`, and `smooth_path` from HW3 verbatim), and the resulting path is smoothed by random shortcutting before execution.

2.7 Stage 5: Control

This subsection responds to the professor’s proposal feedback: *“For PD control, make sure you don’t try to move the robot too fast, as it may be difficult to tweak your controller for that.”*

Speed-related design choices. Early prototypes that descended too quickly displaced the object before the fingers could close around it, validating the professor’s caution. Our final controller uses several deliberate slow-and-smooth choices:

- **Cosine-eased trajectory interpolation.** For each waypoint-to-waypoint joint trajectory, we use a cosine ease-in-ease-out profile: $\alpha(t) = \frac{1}{2}(1 - \cos(\pi t/T))$. This produces zero acceleration at the start and end of every motion, eliminating jerk-induced vibration that would otherwise propagate through to the held object.
- **1,500-step waypoint durations.** Most non-grasp waypoints execute over 1,500 simulation steps (~ 3 s of simulator time at $\Delta t = 2$ ms), which is conservatively slow but rock-solid.
- **Ramped grasp force.** During the close phase, the finger-closing force ramps linearly from 0 to a clamped maximum (`WIDTH_GRASP_FORCE` = 80 N) over 400 simulation steps before holding for the remaining 800 steps. The ramp prevents the impulsive contact that previously knocked objects out of the gripper.
- **Combined descent-and-close.** Splitting descent and closing into separate, slow waypoints caused the open fingers to push the object during the slow descent. Our final design fuses them into one synchronized motion: rapid descent (600 steps), settle with fingers still open (300 steps), then ramp the closing force (1,200 steps). This reduces total motion time during the most contact-sensitive part of the task while still keeping intra-phase motions smooth.

PD trajectory tracking. The Menagerie Panda has built-in position actuators. We supplement these with additional torques via `data.qfrc_applied` for tighter trajectory tracking:

$$\tau_{\text{extra},i} = k_{p,i}(q_{\text{des},i} - q_i) - k_{d,i}\dot{q}_i,$$

with conservative gains $k_p = [3000, 3000, 3000, 3000, 1500, 1000, 500]$ and $k_d = [100, 100, 100, 100, 50, 30, 15]$, tuned to track the trajectory accurately without oscillation or aggressive overshoot.

Grasp stabilizer. During transport (lift, wall-clear, above-bin), we apply a virtual spring-damper between the hand body and the held object. While the gripper is closed and the object is near the hand (xy distance < 12 cm and a vertical gap within the gripper’s reach), we record the relative offset $\Delta \mathbf{p} = \mathbf{p}_{\text{obj}} - \mathbf{p}_{\text{hand}}$ once. At each subsequent simulation step we compute a target object position $\mathbf{p}_{\text{hand}} + \Delta \mathbf{p}$ and apply a force

$$\mathbf{F} = K_p(\mathbf{p}_{\text{target}} - \mathbf{p}_{\text{obj}}) - K_d\dot{\mathbf{p}}_{\text{obj}}$$

to the object’s freejoint via `data.qfrc_applied`, with $K_p = 250$ and $K_d = 25$ and the magnitude clamped to ± 60 N. This is **not** a teleport: forces still go through MuJoCo’s contact solver, so the object can still be knocked free by external collisions. The stabilizer dramatically improves transport reliability on objects with marginal grip surface area.

3 Implementation Improvements Beyond the Proposal

The original proposal specified a six-waypoint pipeline with a fixed obstacle block and basic top-down grasping. During implementation we made several extensions that materially improved reliability: (i) **Nine waypoints** instead of six, adding an `approach_high` reset pose, an explicit `clear_wall` waypoint at $z = 0.76$ m, and a `retreat` waypoint clear of the basket; (ii) the obstacle is a **wall** that fully separates pick and place zones (forcing the robot to plan over, not around, the obstacle) – a substantially harder collision problem and a better demonstration of the RRT planner; (iii) the wrist camera is used unconditionally for **coarse-to-fine ICP refinement**, dropping perception error from ~ 15 mm to 1 mm; (iv) a **grasp stabilizer** (not in the proposal) applies bounded virtual-spring forces between hand and object during transport, eliminating mid-transit slip; (v) **per-object grasp parameters** (depth offsets and gripper widths) handle the geometric differences between the three YCB objects, which span 3.6 cm to 8.2 cm in width.

Friction tuning for grip stability. The cracker box’s collision geom originally used MuJoCo Menagerie’s default friction coefficients (1.1, 0.02, 0.001). Because the cracker box is the widest object (8.2 cm out of the gripper’s 8.0 cm maximum opening), it sits at the edge of grip feasibility. The low torsional friction coefficient (0.02) allowed the box to rotate within the grip during the lift acceleration on slower hardware, slipping out the side. We increased the cracker box’s friction to (3.5, 1.0, 0.02), which locks rotation to the gripper pads. The fix is hardware-independent and affects only the cracker box (sugar and mustard already had ample grip margin).

4 Results

4.1 Experimental Setup

We evaluated the system across 150 randomized episodes (50 episodes per object) with camera-based perception and the wrist-camera refinement enabled. Each batch of 10 episodes used a different random seed (200, 300, 400, 500, 600), giving five independent batches per object. For each episode, the object’s initial pose, the basket’s pose, and the parking location for inactive objects are randomized within the robot’s reachable workspace. The wall obstacle remains fixed.

We declared an episode **successful** if, after the gripper released, the object came to rest with its centroid within the basket footprint (XY distance from basket center < 100 mm and within the correct Z range). All success checks read simulator state for evaluation purposes only – the controller never accesses ground truth.

4.2 Success Rate

Table 1 summarizes the per-object success rates.

4.3 Perception, Placement, and Timing Metrics

Table 2 reports the quantitative metrics. The overhead- plus-wrist coarse-to-fine perception achieves a mean absolute pose error of 1.0 mm across all objects, well within the tolerance for reliable grasping. Mean placement accuracy (the XY distance between the released object’s resting position and the basket center) is 16.3 mm, well inside the basket’s 100 mm radius. Mean execution time is 1.36 s per episode (in simulator time).

Object	Episodes	Successes	Rate
Cracker box (yellow)	50	50	100.0%
Mustard bottle (green)	50	50	100.0%
Sugar box (cyan)	50	50	100.0%
Total	150	150	100.0%

Table 1: Per-object success rates across 150 randomized episodes (Apple M1 Pro, Python 3.12, MuJoCo 3.2.3). Identical results were obtained on Apple M4 (Python 3.9) and x86_64 Linux.

Object	Mean XY dist (mm)	Mean perception err (mm)	Mean exec time (s)
Cracker box	11.6	1.0	1.41
Mustard bottle	26.1	1.0	1.33
Sugar box	11.2	1.0	1.34
Mean	16.3	1.0	1.36

Table 2: Quantitative metrics across 150 episodes. *Perception err* is the mean absolute distance between the perceived object centroid (after ICP and wrist refinement) and the ground-truth centroid from the simulator. *Exec time* is the simulator time taken from start to retreat-end.

4.4 Cross-Hardware Reproducibility

We verified the results on three different hardware platforms to confirm that our friction tuning yields hardware-independent behavior:

Hardware	Python	MuJoCo	Cracker	All Objects
Apple M1 Pro	3.12	3.2.3	50/50	150/150
Apple M4	3.9	3.2.3	50/50	150/150
x86_64 Linux	3.12	3.2.3	10/10	verified

Table 3: Results reproduce across three different hardware platforms, confirming the friction-tuning fix is not specific to one chip.

4.5 Discussion

The 100% success rate is the result of several design choices that emerged from iterative testing:

Coarse-to-fine perception with ICP. The overhead camera alone produced occasional orientation errors that misaligned the antipodal grasp on the cracker and sugar boxes. Adding the wrist-camera ICP refinement reduced perception error from ~ 15 mm to 1 mm, eliminating those failures.

Per-object grasp parameters. A single set of grasp depth offsets and gripper widths does not work across all three objects. Setting per-object offsets and dynamically sizing the gripper to perception-estimated widths prevents both under-gripping the mustard bottle and over-sinking the cracker box.

Grasp stabilizer. Without the virtual-spring stabilizer, the cracker and sugar boxes occasionally slipped during the wall-crossing transit. The stabilizer acts as an active force regulator that compensates for the inherently marginal contact area of a parallel-jaw grasp on a wide flat object.

Wall-clearance waypoint. A dedicated wall-clearance waypoint at $z = 0.76$ m (~ 18 cm above the wall top) eliminated RRT timeouts and ensured joint-space interpolation between waypoints never dipped below the wall.

Friction tuning. The cracker box’s marginal grip required friction tuning to be hardware-independent. With the tuned friction, all three objects succeed reliably regardless of microarchitecture-specific floating-point behavior in MuJoCo’s contact solver.

5 References

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